

Questions of regression Analysis

Question 1

A retail chain wants to analyze the relationship between **customer footfall** and **daily sales**. The following data were collected for 10 days:

Footfall (in hundreds)	15	18	20	22	25	27	28	30	32	35
Sales (in thousands)	120	140	160	180	210	230	250	270	300	320

- Fit the regression model sale on Footfall also interpret the result.
- Estimate the true value of slope & intercept using 90% Confidence interval.
- Test the significance of the slope of the regression line & intercept.
- Predict daily sales when footfall is 24.
- Calculate the correlation coefficient and interpret the result.

Question 2

A manufacturer wants to investigate the relationship between **production cost** and **profit earned**. The data for 10 weeks are:

Production Cost (in thousands)	80	85	90	95	100	105	110	115	120	125
Profit (in thousands)	150	165	180	195	210	225	240	260	275	290

- Fit the regression model Profit on Production cost also interpret the result.

- b) Estimate the true value of slope & intercept using 90% Confidence interval.
- c) Test the significance of the slope of the regression line & intercept.
- d) Predict the profit when production cost is 92.
- e) Calculate the correlation coefficient and interpret the result also find coefficient of determination.

Question 3

A transportation company wants to know if there is a relationship between **fuel consumption** and **distance traveled** by its delivery trucks. Data collected for 10 trips are:

Fuel Used (liters)	20	22	25	27	30	32	34	36	38	40
Distance (km)	180	200	230	250	275	295	310	330	350	370

- I. Fit the regression model Distance on Fuel usage also interpret the result.
- II. Estimate the true value of slope & intercept using 90% Confidence interval.
- III. Test the significance of the slope of the regression line & intercept.
- IV. Predict the distance when fuel used is 28 liters
- V. Calculate the correlation coefficient and interpret the result also find coefficient of determination.

Question 4

A university is studying the connection between **study hours** and **exam scores** among 10 students:

Study Hours	4	5	6	7	8	9	10	11	12	13
Score	55	60	65	68	72	78	82	86	90	94

- a) Fit the regression model Score on Student Study also interpret the result.
- b) Estimate the true value of slope & intercept using 90% Confidence interval.
- c) Test the significance of the slope of the regression line & intercept.

- d) Predict the score if a student studies 9.5 hours.
- e) Calculate the correlation coefficient and interpret the result also find coefficient of determination.

Question 5

A farm wants to study the relationship between **fertilizer usage** and **crop yield** across 10 plots:

Fertilizer (kg)	10	12	14	16	18	20	22	24	26	28
Yield (kg)	90	110	130	150	170	190	210	230	255	275

- I. Fit the regression model crop yield on fertilizer also interpret the result.
- II. Estimate the true value of slope & intercept using 90% Confidence interval.
- III. Test the significance of the slope of the regression line & intercept.
- IV. Predict crop yield when fertilizer used is 17 kg.
- V. Calculate the correlation coefficient and interpret the result also find coefficient of determination.